



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

AMS 210

Applied Linear Algebra

Unit 0

Course Organization

Unit Goals

- **Present the structure of AMS210 course.**
- **Outline the assessments for the course.**
- **Convey the expectations for homework submissions.**
- **Understand the important of regular attendance.**
- **Learn the best approach to master the content.**

Course structure.

- The course is taught in-person on site by an instructor.
- The content of this course is provided by Prof. David Green from Stony Brook University.
- Students can access the course materials via Teams, with access granted via an access code in the first class.

The teaching team

Instructor:

- Dr. Francis Ting toting@ahu.edu.cn

Teaching Assistants:

- Mr. Man Yan
- Ms. Zhang Wannan

Textbooks

- **Primary Class Textbook:**

Introduction to Linear Algebra, 5th edition.

Gilbert Strang (2016).

- **Optional Supplementary Textbook (free download).**

Introduction to Linear Algebra: Models, Methods and Theory.

Alan Tucker (1995).

Download: <https://commons.library.stonybrook.edu/ams-books/1/>

Evaluation

Homework:

40% of final grade; 8 homework assignments worth 5% each.

Quiz:

40% of final grade; 4-6 quizzes.

Exams:

20% of final grade.

Teaching philosophy

- My goal in teaching this (and every) course is to guide students toward developing **a real understanding** of the material.
- My pedagogical choices are based on achieving this **for all students who wish to succeed**.
- I understand that people have different learning styles, and I do my best to accommodate this.
- Additionally, I understand that students have a varying comfort levels with mathematics, and I am here to improve their understanding and skills.

Teaching philosophy

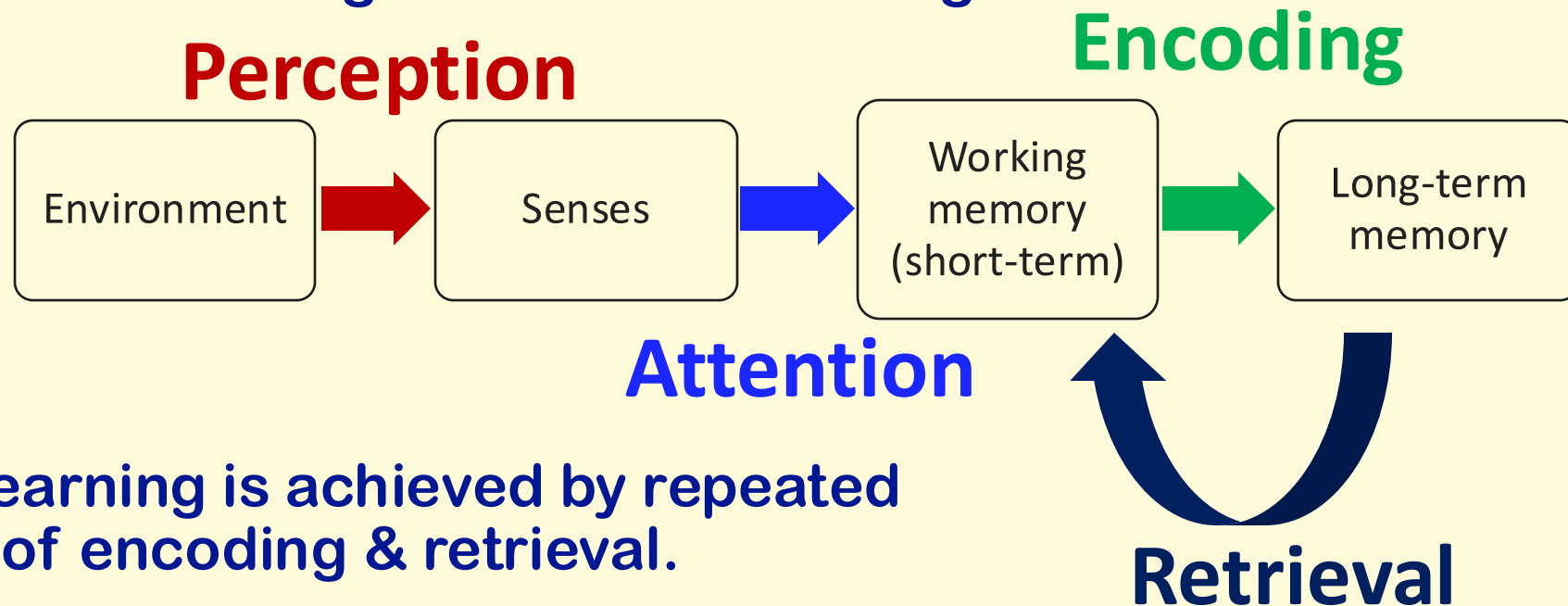
- There are various resources available to support learning the material:
 - Attending (and paying attention to) lectures.
 - Reading the textbook before/after lectures.
 - Completing homework and submit before the deadline.
 - Reviewing solutions to homework problems.
 - View relevant videos and try some optional problems.

Teaching philosophy

- There are several levels of knowledge and application that we will explore throughout the class:
 - Understanding and explaining underlying concepts.
 - Executing the fundamental mechanics of key techniques.
 - Solving a mathematical problem by identifying the technique(s) to use, then executing the mechanics.
 - Solving a “real-world” type problem by formulating a mathematical model, identifying the technique(s) to use, executing the mechanics, and interpreting the result.
- We will explore all these levels for each topic we consider.

Some thoughts on effective learning

- Part of the neurological basis of learning:

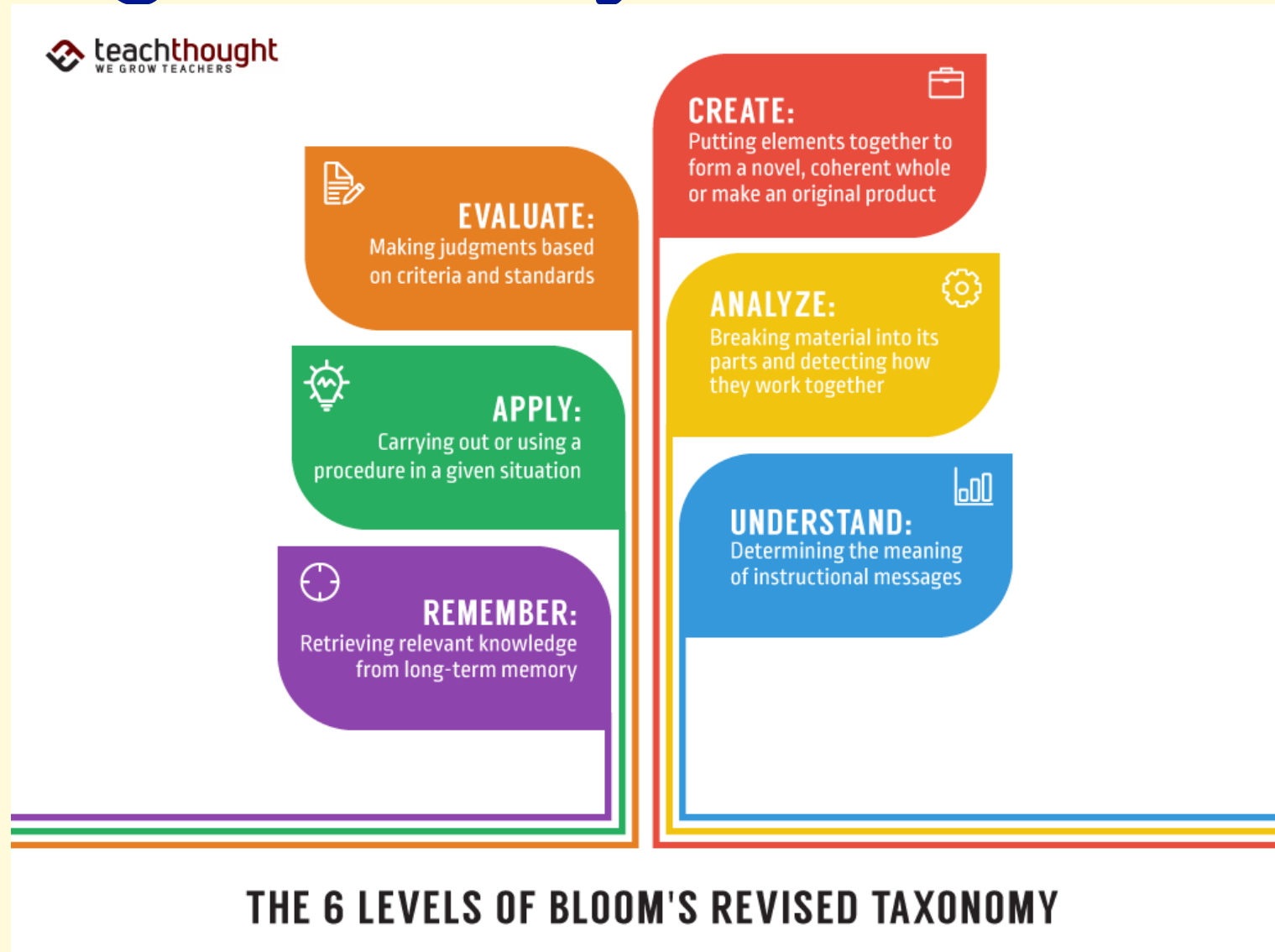


- Deep learning is achieved by repeated cycles of encoding & retrieval.
- Perception and attention are necessary, but not sufficient.

Some thoughts on effective learning

- **Perception:**
 - Sitting through lecture and looking through the book.
- **Attention:**
 - **Carefully** listening to lecture and reading the book.
- **Encoding:**
 - Working through problems, thinking about and discussing the material.
- **Retrieval:**
 - Taking tests, doing “reference-free” problems, teaching and explaining to others.

Learning has many levels.



Learning has many levels.

Level	Abilities
Novice	• Missing key vocabulary or syntax that make it difficult or impossible to follow a written or oral presentation of the material
Beginner	• Ability to follow a written or oral presentation with good comprehension.
Advanced Beginner	• Able to reproduce the solution to a similar problem, using an example solution as a guide.
Competence	• Able to solve a problem of a type that you have seen before, without referring to an example solution. • Able to explain the reasoning underlying a solution to a problem of a type seen before and to identify any flaws in a presented solution.
Proficiency and Mastery	• Able to solve a problem of type that you have not seen before (perhaps using references). • Able to explain the reasoning underlying a solution to a problem of a new type and to identify any flaws in such a solution.

Learning has many levels.

I encourage everyone to be aware of what they are learning and how deeply they know the material. Part of this involves developing an accurate assessment of your confidence in your response to a question:

1. Do you understand completely what a question is asking, or are there terms or symbols that you don't fully understand?
2. Do you know the correct approach to take in solving a problem, do you think you can make a good guess, or do you have no idea where to start?
3. Are you confident in your ability to execute on the correct approach (barring the occasional "silly" mistake), or are there holes in your recall of the mechanics?
4. Do you have enough of an understanding that you can explain the significance of your answer and/or identify inconsistencies that may be a sign of an error in calculation?

Core Homework

- Each homework contains a set of **10 core problems**
- Homework will be graded partially based on completeness, and partly on correctness of your answer. You must show all calculation steps towards the answer to receive full credit.
- Due to limited resources, feedback on your homework will largely be limited to your numerical grade; you will not be given detailed explanations of your error(s).
- Complete solutions will be posted, and it is your responsibility to check your work against the posted solutions.

Optional Homework Problems

- Each homework includes a set of **optional** pre-homework practice problems.
 - These are intended for students who need extra practice prior to tackling the main assignment.
- Each homework also includes a set of **optional** challenge problems.
 - These are intended for students that wish to push their understanding of the material beyond the core foundations and are recommended for students that are considering pursuing more advanced studies in Applied Math.

What is allowed in solving homework?

- Everything needed to solve the homework problems will be provided in lecture and/or in the class texts.
- However, you are welcome to use other texts, watch additional tutorial videos, and look for guidance online.
- You may NOT make use of any “Solutions Manual” or other published set of solutions to the problems!
- You may NOT post questions online directly asking for solutions to any problems.
- **ChatGPT:** Use this as a tool for you to learn effectively and not as a tool to cheat! Always make sure that you can work out a problem on your own to verify your understanding and mastery of the content.

Collaboration on homework

- You are welcome to discuss homework problems with your classmates; **collaborative learning** can be highly effective.
- However, your solutions should not be similar as you are expected to write the solution **independently** by yourself.
- Allowed collaboration includes talking to each other about the general approach to solving a problem, sharing resources that helped in your understanding, and **cross-checking your final answers** with each other.
- You should NOT copy solutions from any source directly. You still do not understand unless you can produce the answer independently by yourself.

My expectations for students.

Make a sincere effort to learn the material:

- Read the relevant sections of the book and watch the lectures before the associated classroom meetings (and before starting homework).
- Start your homework assignments well in advance of the due date.
- Review posted homework solutions promptly and identify where you made mistakes.
- Ask for help if you are having trouble!!!
- **Be honest in your work!**